

FPGA based Real-Time simulation of FlyBack converter using graphical programming tools

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Abstract — This paper presents a novel FPGA Real-Time simulation solution based on modern graphical programming and automated testing tools. The proposed setup was used for simulation of high-speed power electronics switching phenomenon within some considered circuit topologies. In the actual setup, the behavior of a power electronic converter stage was hardware – emulated using a powerful FPGA (Field Programmable Gate-Array) logic processing unit. This logic – processor unit was used to run fast, and intensive computing tasks related to the desired power electronics circuit topology behavioral simulation techniques. The FPGA unit was equipped with analog and digital input and output ports, so that the resultant digital real-time simulation waveforms were physically generated as variable voltage signals. The real variable voltage signals required in the control scheme of the considered power electronics topology were also acquired using the built-in digital and analog ports. The physical control variable voltage signal was generated using a digital control law algorithm implemented within an embedded system process controller (NI MyRIO). This kind of simulation approach is so-called Hardware in the Loop (HIL) Real-Time simulation. This simulation technique requires two real-time computing systems, the HIL simulation computer (NI PXI) and the embedded controller (NI MyRIO). As software infrastructure, specialized real-time simulation and automated testing software environments were used. The whole presented HIL solution is mainly centered on the National Instruments devices plus the LabVIEW software infrastructure and also the VeriStand test automation environment including the OPAL-RT eHS circuit solver Add-on. Matlab – Simulink was also used to implement the power electronics circuit model. The overall system represents a Hardware In the Loop and Virtual Instrumentation Engineering Workbench solution.

Keywords — *FPGA power electronics, eHS solver, OPAL-RT, PXI, VeriStand, LabVIEW, Matlab-Simulink*

I. INTRODUCTION

The process of power electronics circuit simulation represents an intensive computing task. It requires fast response of the considered processing system in order to achieve the real-time response [1]. Ordinary personal computers or specialized simulation workstations can partially handle this task by using multi-core processor, and multi thread processing [2], [3]. The most advanced simulation-ready workstations can solve the computing task by passing some of the intensive processing operations to the GPU (Graphical Processing Unit) based simulation (e.g. Nvidia CUDA core processing) [4], [5], [6], [7]. Nowadays, special processing and computing units have been developed to serve

artificial intelligence tasks - tensor processing units (TPU) [8] and they are meant to solve complex AI algorithms and to execute repeatedly operations and iterations of a considered computing technique [9]. The FPGA use is the most appropriate approach to simulate a power electronics converter topology at *high-speed, high switching frequency, with high fidelity and low latency under real-time response conditions*. On the other side, the only way to simulate the circuit on a workstation or personal computer (with decent computing resources and relatively good fidelity) is to do a so-called “offline” simulation. Offline simulations allow the computing unit to perform the simulation step by step. At the end of a computation step (within the simulation process) the results are represented punctually. Sometimes, the results are represented only at the end of the simulation as a resultant waveform or numeric array. The best sample-time that personal computers or workstations can achieve is 10^{-7} [s]. But within that sample period, the results are not represented in real-time. The sample period parameter in this condition represents only the results accuracy and simulation fidelity. With some intensive processing efforts 10^{-8} [s] can be achieved with some powerful processors [10], [11], [12].

The solution in this case consists in the use of an FPGA logic processor as simulation accelerator, because it can handle a greater volume of data [13]. The internal architecture of such a logic processor can be reconfigured and repartitioned according to the end-user demand. Every newly created portion of the FPGA can manage a separate logic computing task. This allows the FPGA to perform fast parallel computing [14].

In power electronics, switching devices and switching operations can be easily simulated by an FPGA logic unit [15]. Logic gates are mainly composed of high-speed switching mode transistors, which can act as the main switching elements in power electronics converters topology models. Examples of commercial HIL simulation systems are listed: Opal-RT – OP HIL series [16], National Instruments PXI HIL [17], Plexim – PLECS RT Box [18], Speedgoat HIL system [19], dSPACE SCALEXIO HIL [20], Typhoon HIL system [21], ETAS – HIL system [22].

Each HIL system manufacturer uses a different software and hardware configuration approach to provide more accurate simulation results in everyday engineering challenges. The ‘problem-solutioning’ mechanism approach provided by the HIL manufacturer does not include only hardware components, it also includes software bundles and routines. As an example, mathematical and numerical

computing procedures implemented as software routine libraries, which are meant to help the end user to find the best results within the simulation process, are called solvers and they are most of the times part of the simulation environment [23]. The HIL systems require an automatic testing software platform or a complex simulation environment software to perform the automatic testing scenarios. There are several dedicated simulation environments available on the market: National Instruments – LabVIEW [24], MathWorks–Matlab–Simulink [25], Plexim–PLECS Standalone and BlockSet [26], Power Sim–pSim [27], Altair–Altair Embed or SolidThinking Embed [28].

Current simulation environment software can also generate embedded code for both real-time processor and real-time FPGA applications, like automated test scenarios or real electrical circuit behavior simulation. LabVIEW, Matlab – Simulink and many other related software solutions are also used for developing digital control algorithm code for embedded controllers (microcontrollers, DSPs, FPGAs) [29]. Most of the HIL solutions come with their own management software, and test automation workbench [30], [31]. The key feature of some HIL solutions software is that the dedicated management software can be coupled with a complex model-based simulation environment (Simulink). It means that the product-specific HIL management solution software has shared compatibility with third party software, allowing data exchange between the target HIL hardware and the complex simulation environment. For example, in power electronics and electrical engineering problems, both specialized HIL management software and electrical circuit and solver routines are required to find the best results. The combination of software solutions like National Instruments VeriStand [32] and Opal-RT eHS Power Electronics Add-on [33] and Matlab–Simulink with SimScape Electrical or SimPowerSystems Specialized [34] is used for this purpose. National Instruments VeriStand is virtual test work-bench software dedicated to National Instruments hardware that can manage all the hardware resources and embedded programming tasks within the NI HIL computing and data acquisition systems. Simulink with SimScape Electrical or SimPowerSystems Specialized is used to generate model-based simulations of the circuits and embedded code for the HIL computing system. In order to develop digital control embedded software for the automated controller, LabVIEW is used.

A FlyBack converter will be simulated on the proposed Real-Time target HIL system for validation of this novel approach.

II. WORKING PRINCIPLES AND CONCEPTS

A. The Opal-RT eHS solver [35]

The eHS Solver uses Modified Nodal Analysis (MNA) [36] to generate a conductance matrix that returns the voltage at each node of a circuit and the current in each branch. The conductance matrix of the circuit is generated independently from the state of the switches, and therefore does not need to be recomputed when a switch is opened or closed during the simulation. This is achieved through the implementation of the Pejovic method. This method represents each switching element as an impedance. A conducting switch is represented as an inductor and an open switch is represented as a capacitor. The inductors and capacitors are further simplified using Backwards Euler [37] discretization method, and their

equivalent circuits become a current source with a shunt resistance.

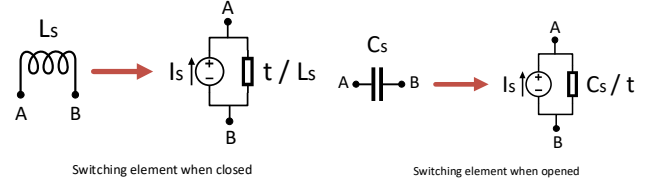


Fig. 1. Switching element states and equivalent circuits

For the conductance matrix to remain constant throughout switching events, the following equations, which define the Switch Conductance “ G_s ”, must remain TRUE.

$$G_s = \frac{t}{L_s} = \frac{C_s}{t} \quad (1)$$

Where t is the simulation time step of the circuit. When building the nodal matrix, G_s is defined for each switch, typically within a range of 0.001 to 10. It determines the value of the inductance and the capacitor representing the switch in conducting and non-conducting states.

$$L_s = \frac{t}{G_s} \quad (2)$$

$$C_s = t \cdot G_s \quad (3)$$

For example, for a time step of 100 ns and a G_s of 1, the switch will be represented by the following inductance and capacitance:

$$L_s = \frac{t}{G_s} = \frac{100}{1} = 100 \text{ [nH]} \quad (4)$$

$$C_s = 100 \cdot 1 = 100 \text{ [nF]} \quad (5)$$

The “ G_s ” value for each switch is dependent on the following factors: the topology of the circuit and the circuit parameters: base power, source current, source voltage, duty cycle, load current, load voltage;

Typically, a good approximation for “ G_s ” is:

$$G_s = \frac{I_{max}}{U_{max}} \quad (6)$$

Where I_{max} is the maximum current flowing outside the converter and U_{max} is the maximum voltage at the arm middle point.

B. The FlyBack converter [38], [39]

The Flyback converter (Fig. 2) is a buck-boost converter with isolation between its input and output. The inductor used in the buck-boost configuration is replaced by a transformer for isolation and storing energy before sending to the output via the output capacitor.

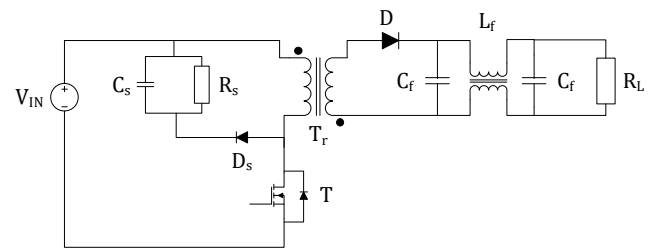


Fig. 2. FlyBack converter schematic

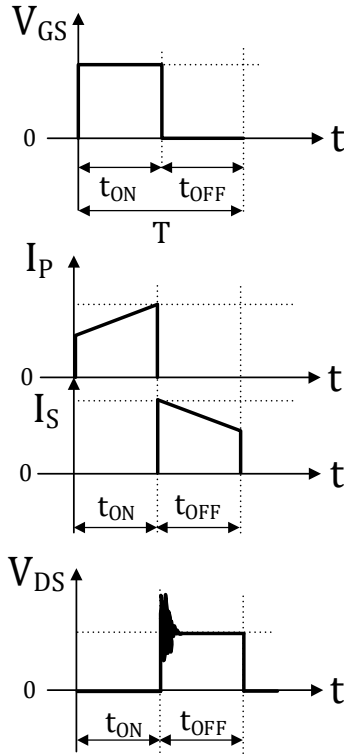


Fig. 3. Working waveforms of a FlyBack converter [40]

The working waveforms are presented in Fig. 3. The operating principle is similar to the one of the buck-boost converter, but an additional transformer is used to achieve galvanic isolation of the input and output. The theoretical transfer function of the flyback converter is:

$$U_{out} = -\frac{N_2}{N_1} \cdot \frac{D}{(1-D)} \cdot U_{in} \quad (7)$$

Where D is the duty cycle, and N_1 and N_2 are the turns numbers of the primary and secondary windings, respectively.

In this paper, a step-down flyback converter was simulated using the proposed Real-Time HIL system.

III. EXPERIMENTAL HARDWARE AND SOFTWARE SETUP

A. Hardware setup description:

The test bench for the Real-Time HIL solution is depicted in Fig. 4, with different color highlights of the components.

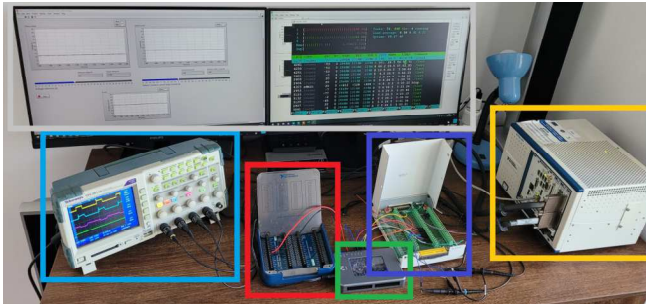


Fig. 4. Hardware setup of the HIL system

The test bench components are being listed below:

- NI PXIe - 1071 chassis (yellow).
- NI PXIe - 8840 Quad - Core PXI controller;
- NI PXIe - 7868R FPGA multi-functional PXI board;

- NI SHC68 - C68 - RDIO2 cable;
- NI SCB - 68 HSDIO terminal block (red);
- NI SHC68 - 68 - RMIO cable;
- NI SCB - 68 - MIO terminal block (purple);
- NI 1900 - MyRIO embedded controller (green);
- Tektronix TPS 2014B digital oscilloscope (blue);
- HP - Z440 desktop computer (gray);

The NI PXIe-7868R is a multi-functional FPGA Reconfigurable I/O Module based on Kintex-7 325T FPGA processing unit [41]. This module is used for computing the power electronics circuit states, and to acquire high speed sampled signals. The approximative minimum admissible sample time of this FPGA unit is 20^{-7} [s], when simulating power electronics circuits using the OPTAL-RT eHS circuit solver. The NI PXIe-7868R features a user-programmable FPGA for high-performance onboard processing and direct control over I/O signals for complete flexibility of timing and synchronization. It provides the resources to design applications that require precise timing, custom protocol communication, sensor simulation and high-speed control. The PXIe-7868 features a dedicated A/D converter (ADC) per channel for independent timing and triggering. This design offers multi-rate sampling and individual channel triggering, which are outside the capabilities of typical data acquisition hardware.



Fig. 5. The NI PXIe-7868 - Reconfigurable FPGA module [42]

The PXIe-1071 (Fig. 5) chassis is designed for a wide range of test and measurement applications and provides a high-bandwidth backplane.

The PXIe 8840 (Fig. 6) is an embedded controller for PXI systems used for processor-intensive, modular instrumentation, and data acquisition applications. It provides two 10/100/1000BASE TX (Gigabit) Ethernet ports, two USB 3.0 ports, four USB 2.0 ports, as well as an integrated hard drive, serial port, and other peripheral I/O. It features, a 2.7 GHz Dual-Core or 2.6 GHz Quad-Core Processor suitable for running an embedded Real-Time ready Operating System (NI Real-Time Linux Image).



Fig. 6. The NI PXIe-1071 chassis [43]



Fig. 7. The NI PXIe 8840 – Embedded controller [44]

B. Software setup description:

The HIL system will be implemented using the considered software suites and solutions:

LabVIEW is a system engineering software for applications that require test, measurement, and control with rapid access to hardware and data insights. The LabVIEW programming environment simplifies hardware integration for engineering applications so that you have a consistent way to acquire data from NI and third-party hardware. It allows data acquisition manipulation, through developed algorithms for data analysis and advanced control with included math and signal processing IP. To ensure compatibility with other engineering tools, LabVIEW can interoperate and reuse libraries from other software and open-source languages.

VeriStand software is intended for real-time test applications, such as stimulus generation, data acquisition, calculated channels and custom channel scaling. VeriStand application software allows the communication with I/O channels, data logging, stimulus generation and host communication for NI real-time hardware. It also provides the possibility to import simulation models and control algorithms, respond to events and enable test automation with macro recording, Test Stand, .NET, and other software. A run-time editable user interface permits interaction with the user. A variety of software environments such as LabVIEW, ANSI C/C++, Python and ASAM XIL can be used to add custom functionality to VeriStand.

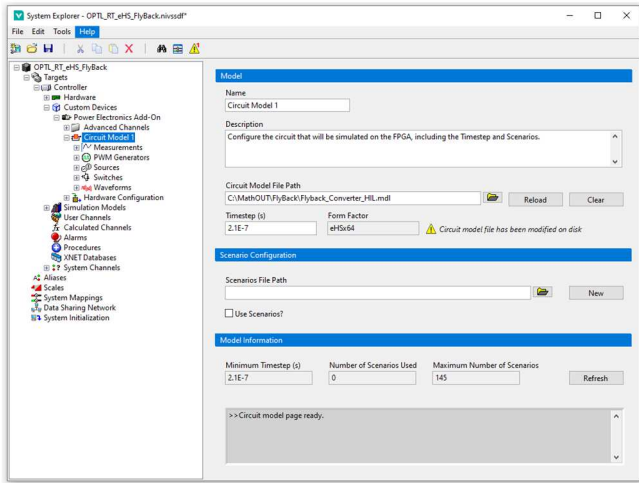


Fig. 8. OPAL-RT FPGA-Based Power Electronics Add-On for VeriStand

of a graphical user interface wizard. The MathWorks SimScape Electrical Toolbox can be used in order to develop the power electronics converter model. The FPGA-Based Power Electronics Add-On for VeriStand also provides user interface wizard for configuration of I/O channels, generation of stimuli, visualization and recording of electric signals, and execution of test scenarios during simulation. eHS is part of the FPGA-Based Power Electronics Add-On for VeriStand and it provides a convenient user interface enabling the import of real-time models created in other simulation tools: SimScape Power System Simulink toolbox, PSIM, PLECS BlockSet, or Multisim.

Matlab-Simulink is a block diagram environment used to design systems with multidomain models, to simulate before transferring it to hardware and deploy without writing code.

The software components involved in the HIL system are:

- NI LabVIEW 2021 – SP1 and MyRIO Toolkit;
- NI VeriStand 2021 R3;
- Opal-RT eHS Power Electronics Add-on for VeriStand;
- Matlab – Simulink R2016b with Sim Power Systems;

IV. IMPLEMENTATION AND SIMULATION RESULTS

In order to validate the functionality of the proposed real-time simulation model through different working scenarios, the HIL setup presented in Fig. 9 was considered.

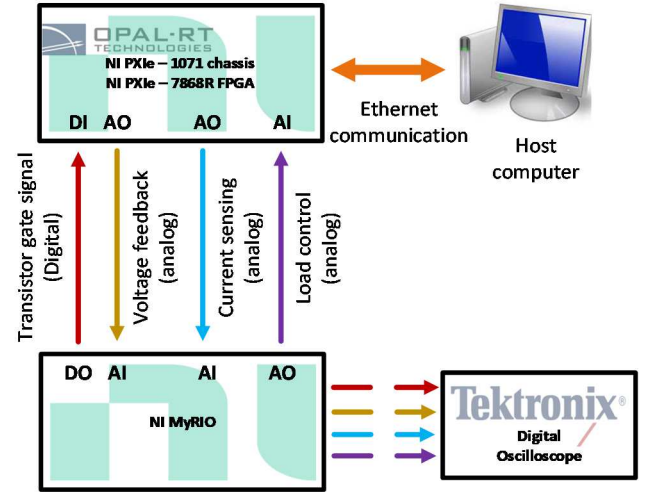


Fig. 9. HIL system architecture bloc diagram

In the actual described HIL system architecture, the NI PXIe-1071 and the NI-7868R FPGA I/O board act as the plant process (the converter model) and MyRIO acts as the embedded digital controller (the control algorithm).

Within VeriStand and the eHS power electronics add-on setup process, the chosen FPGA hardware configuration was the “eHSx64_Dual_PMSM_VDQ_IO_7868R” FPGA bitstream file [45]. The preset configurations enables different features regarding the hardware capabilities, according to the user’s need. The I/O ports were mapped for the HIL computing system, as presented in Table I – for PXI and Table II – for the MyRIO embedded controller.

TABLE I. PXI I/O PORTS MAPPING IN VERIStand AND P.E. ADD-ON

Port Symbol	PXIe - 7868R I/O ports mapping	
	I/O port type and mode	Port role in model
AI_0	Analog input	Load control
AO_0	Analog output	Output voltage feedback

Port Symbol	PXIe - 7868R I/O ports mapping	
	I/O port type and mode	Port role in model
AO_1	Analog output	Output voltage feedback (scope)
AO_2	Analog output	Input current monitor (scope)
AO_3	Analog output	Diode current monitor (scope)
AO_4	Analog output	Output current feedback
DO_0	Digital output	Gate signal monitor (scope)
DI_27	Digital input	Transistor gate control signal

TABLE II. PXI I/O PORTS MAPPING FOR MYRIO

Port Symbol	MyRIO I/O ports mapping	
	I/O port type and mode	Port role in model
AI_0	Analog input	Output voltage feedback
AI_1	Analog input	Output current feedback
AO_0	Analog output	Load control
DO_3	Digital Output	Transistor gate signal control

The software setup for this model consists in a Simulink model of the FlyBack converter, a VeriStand project and two LabVIEW projects. The first LabVIEW model provides open-loop functionality for the FlyBack converter and the other model is used for the closed-loop control of the FlyBack converter model. The simulation parameters for the converter's model within Simulink are presented in Table III.

TABLE III. SIMULINK MODEL PARAMETERS

Parameter symbol	Simulink simulation model parameters	
	Parameter name	Parameter value
Ts	Effective sample time	1e-6 [s]
T	Total simulation time	1 [s]
Uin	Input voltage	150 [V]
rs	Power supply series resistance	0.05 [Ω]
Cs	Snubber capacitor	50 [μ F]
Rs	Snubber resistor	10 [k Ω]
Ls	Snubber inductor	1 [μ H]
Rs DIODE	ON state resistance of DIODE	0.001 [Ω]
Rs DIODE	Snubber resistor DIODE	500 [Ω]
Cs DIODE	Snubber capacitor DIODE	250 [pF]
Ron IGBT	ON state resistance of IGBT	0.001 [Ω]
Rs IGBT	Snubber resistor IGBT	100 [k Ω]
Cs IGBT	Snubber capacitor IGBT	Infinity [F]
Pn	Nominal transformer power	25 [W]
fn	Nominal transformer frequency	10 [kHz]
V1	Nominal primary winding voltage	150 [V]
R1	Nominal primary resistance	0.15 [Ω]
L1	Nominal primary inductivity	30 [μ H]
V2	Nominal secondary voltage	30 [V]
R2	Nominal secondary resistance	0.015 [Ω]
L2	Nominal secondary inductivity	0 [μ H]
Rm	Magnetization resistance	100 [k Ω]
Lm	Magnetization inductivity	10 [mH]
C	Output voltage filtering capacity	800 [μ F]
Rc	Capacitor armature resistance	0.05 [Ω]
I1	Initial output current	0.05 [A]
rp	Power supply parallel resistance	100 [k Ω]
fsw	Switching frequency	10 [kHz]
n	Turns ration	10:1

The eHS solver requires special naming for the power supplies, measurement means and switching elements within the SimPowerSystems model. To import the Simulink model into VeriStand and Opal-RT's eHS Power Electronics Add-on, the following naming and mappings have been done in the SimScape SimPowerSystems model:

TABLE IV. MODEL COMPONENTS MAPPING IN SIMULINK

Model Symbol	Simulink FlyBack model components	
	Component type	Component role
Y01 V1	Voltage sensor	Input voltage sensing
Y02 A1	Current sensor	Input current sensing
Y03 V2	Voltage sensor	IGBT voltage drop sensing

Model Symbol	Simulink FlyBack model components	
	Component type	Component role
Y04 A2	Current sensor	Diode current sensing
Y05 A3	Current sensor	Output current sensing
Y06 V3	Voltage sensor	Output voltage sensing
U0 V1	Controlled voltage source	Input voltage supply
U1 I1	Controlled current source	Load current control
SW 01	IGBT switching element	Main switching element
SW 02	Snubber diode	Over-voltage suppression
SW 03	Fast rectifying diode	Output voltage rectifier

The same mappings have been done in VeriStand eHS Power Electronics Add-on in order to match the model components with the circuit solver parameters. On the VeriStand side, the model components were associated with physical hardware outputs such as digital I/Os or analog I/Os.

TABLE V. MODEL COMPONENTS MAPPING IN VERISTAND

Model Symbol	VeriStand FlyBack model components	
	Component name	FPGA assignment
Y01 V1	Voltage sensor	Waveform monitor Wm 1
Y02 A1	Current sensor	Analog output 2 + Wm 2
Y03 V2	Voltage sensor	Waveform monitor Wm 3
Y04 A2	Current sensor	Analog output 2 + Wm 4
Y05 A3	Current sensor	Analog output 4 + Wm 5
Y06 V3	Voltage sensor	Analog output 0 + Wm 6
U0 V1	Controlled voltage source	CPU (VeriStand)
U1 I1	Controlled current source	Analog input 0
SW 01	IGBT switching element	Digital input 27
SW 02	Snubber diode	CPU (VeriStand)
SW 03	Fast rectifying diode	CPU (VeriStand)

After the hardware and software parametrization step, the Real-Time model testing steps were performed. The first test was performed in open-loop conditions, and the second test was performed in closed-loop control conditions. The following settings were made for the open loop operation mode: input voltage 150 [V], output voltage co-related with duty cycle 12 [V], duty cycle 15.89 [%], load current range 0.05–3 [A], PWM switching frequency 10 [kHz]. Both operation modes were implemented within MyRIO controller.

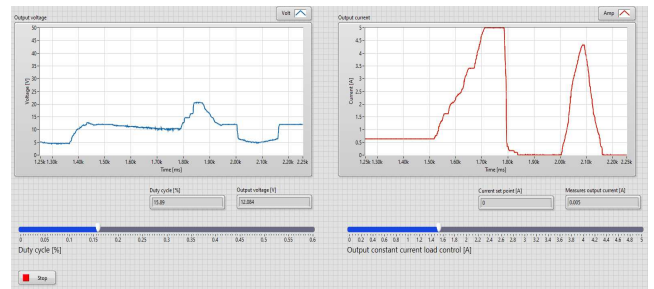


Fig. 10. LabVIEW open loop working mode VI (front panel)

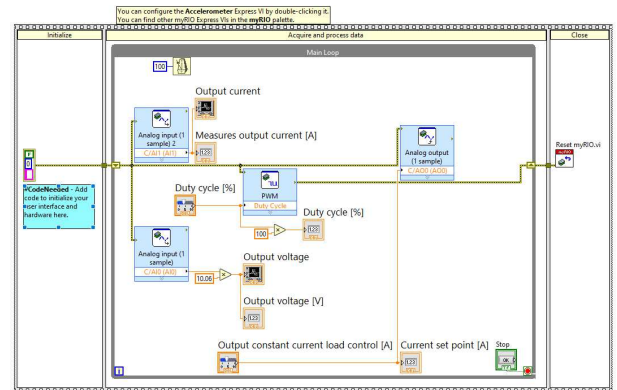


Fig. 11. LabVIEW open loop working mode VI (block diagram)

In the open loop working mode, a fixed duty cycle corresponding to 12 [V] output voltage was set. The considered FlyBack topology is a step-down dc-dc converter, from 150 [V] to 12 [V]. The output current is modified from 0.05 to 3 [A]. The result reflects the fact that the FlyBack converter will not be able to maintain the output voltage level constant at 12 [V] if the load current demand is modified.

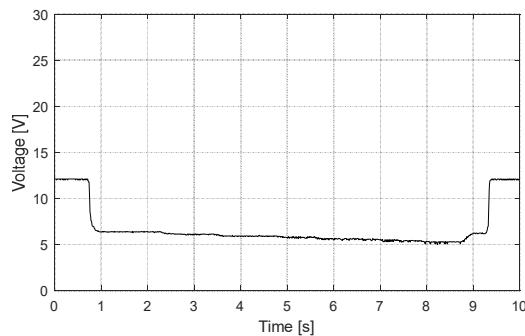


Fig. 12. Output voltage in open loop working mode

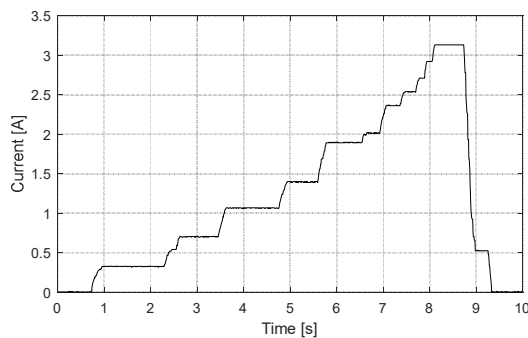


Fig. 13. Output load current variation over time

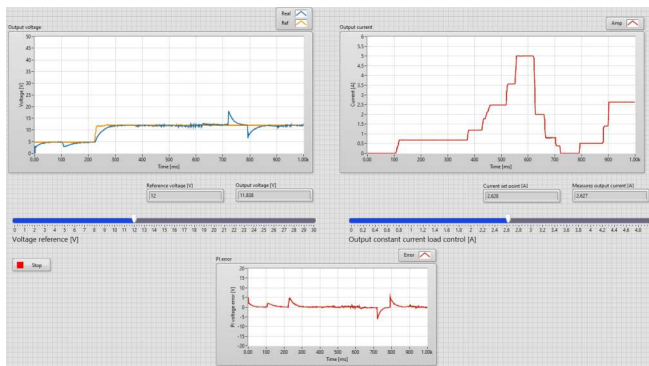


Fig. 14. LabVIEW closed loop working mode VI (front panel)

For the closed loop operation mode, the following test parameters were set: input voltage 150 [V], output voltage 12 [V], load current range 0.05 – 3 [A], PWM switching frequency 10 [kHz].

In the closed loop mode, the reference voltage level is set to 12 [V]. A proportional – integral controller (PI) will adjust the duty cycle automatically in such way that the output voltage level remains the same as the specified voltage reference. The feedback measured output voltage will match the specified voltage reference level regardless of the load current variation. The output current will be modified from 0.05 to 5 [A]. In this case, the result reflects the fact that the FlyBack converter can maintain the output voltage level constant at 12 [V] if the load current demand is modified.

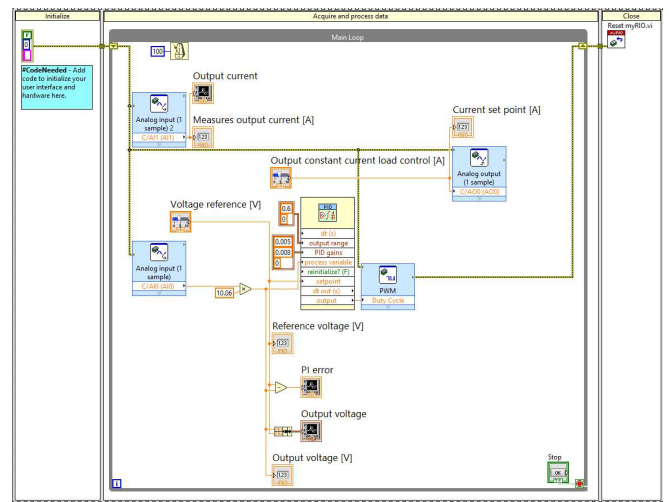


Fig. 15. LabVIEW closed loop working mode VI (block diagram)

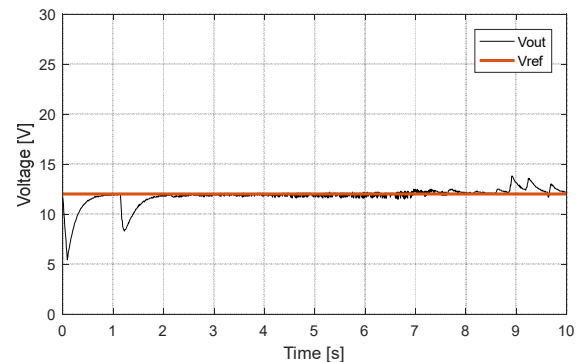


Fig. 16. Output voltage in closed loop working mode

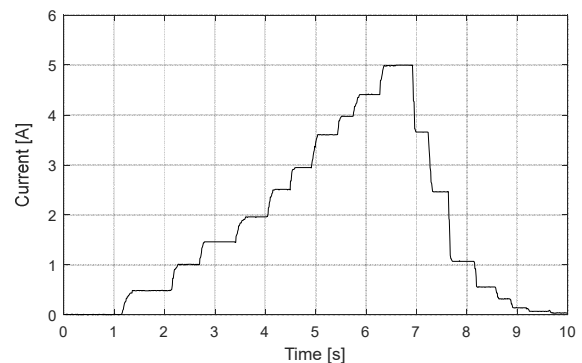


Fig. 17. Output load current variation over time

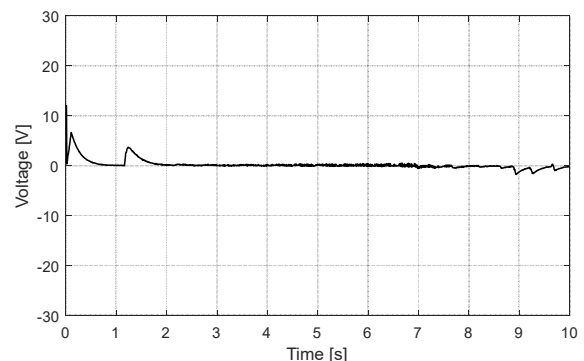


Fig. 18. Difference between output voltage reference level and measured output voltage level (closed loop system error)

The difference between the reference voltage and the measured output voltage level (closed loop system error) tends to be convergent to zero at a certain stability point, as quickly as possible. The duty cycle of the FlyBack converter will be automatically adjusted by the PI controller implemented within MyRIO embedded controller.

The presented results prove the fact that the Opal-RT eHS solver for VeriStand is efficient enough to perform high speed real-time simulation of a switching circuit topology. By performing Hardware in The Loop simulation, the real behavior of the “virtual” circuit topology can be emulated using the FPGA I/O digital and analog ports. The resultant variable voltage signals can be observed with a monitoring setup consisting of a digital oscilloscope attached to the analog output ports of the PXIe-7868R FPGA multifunctional card.

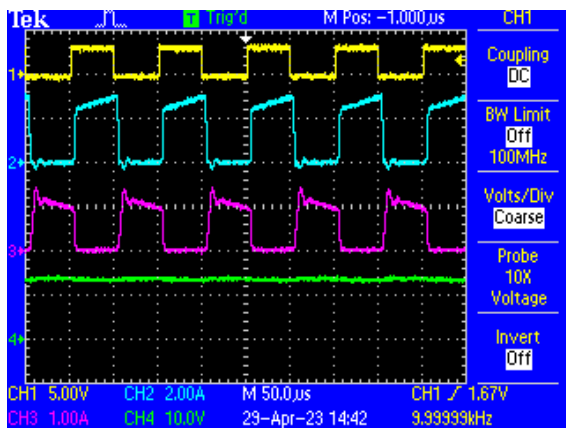


Fig. 19. Digital and analog output voltage and current signal monitoring

Fig. 19 represents the real physical resultant output signals from the HIL simulation. The waveform representation was done using the Tektronix TPS 2014B – digital oscilloscope. The yellow-colored waveform represents the transistor gate signal. The cyan-colored waveform represents the current that passes through the transistor (or the converter input current). The magenta-colored waveform represents the current that passes through the output diode of the FlyBack topology. The green-colored waveform represents the output voltage.

V. CONCLUSIONS AND FUTURE WORK

The present paper proposes a novel HIL setup suitable for didactic purposes in Electrical Engineering domain, in terms of Power Electronics, Electrical Drives and basic passive circuit theory. It also represents a good tool for research and development in high power electrical engineering, electrical grid-related phenomenon emulation and electrical machines emulation with magnetic circuit modeling capabilities. The flexibility and reconfigurability of the model are provided by the large number of I/O ports and the ability of coupling multiple FPGA modular boards inside the PXI chassis, making it suitable for didactic implementations.

The NI PXI hardware can be used as a Power Electronics Real-Time HIL simulation platform, with Opal-RT's eHS Power Electronics Add-on being installed within VeriStand. The embedded controller which runs the control-law algorithm consists of a MyRIO embedded system, programmed with LabVIEW. The VeriStand program-model

is implemented graphically by using Matlab–Simulink SimPowerSystems circuitry toolbox.

The concerns about the current HIL setup are related to the cost of the hardware components and software bundles.

Future research will be focused on highlighting another key aspect of the presented setup, by removing the need for an external embedded controller device (like MyRIO). This can be done due to the interconnectivity of the components: the control Simulink model can be implemented on the PXI's main embedded controller CPU, the NI PXIe-8840 controller features a Quad-Core CPU that can handle the VeriStand model, and the FPGA logic processor can co-operate with the CPU via the dedicated FPGA to CPU AXI link - interface.

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